CMPG 325

INDIVIDUAL SEMESTER PROJECT

MILESTONE 1 — CLIENT DESIGN REVIEW

Project ID	CMPG325-2026-087
Client ID	CLI-087
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Client	Kgomotso Marketing & Advertising Agency
Location	Rustenburg, North West, South Africa
Addressing Block	10.36.0.0/16
Technical Challenge	Network Troubleshooting — Fault Isolation Scenario
Constraint	Limited after-hours wireless access for cleaning and security contractors
CR9	CR9 — Secure remote management access for one off-site administrator

1. Executive Summary

Kgomotso Marketing & Advertising Agency is the client for this CMPG 325 network design project. Milestone 1 establishes the approved-design baseline for later Cisco Packet Tracer implementation. The proposed design uses the assigned 10.36.0.0/16 block and addresses the contractor wireless constraint, CR9 secure remote management requirement, and assigned fault-isolation challenge.

2. Requirements and Scope

Client: Kgomotso Marketing & Advertising Agency

Location: Rustenburg, North West, South Africa

Industry: Professional Services

Addressing: 10.36.0.0/16

Technical challenge: Network Troubleshooting — Fault Isolation Scenario

Constraint: Limited after-hours wireless access for cleaning and security contractors

CR9: CR9 — Secure remote management access for one off-site administrator

Milestone 1 deliverables: Client requirements; physical topology; logical topology; IP addressing plan; initial GitHub repository

3. Physical and Logical Topology

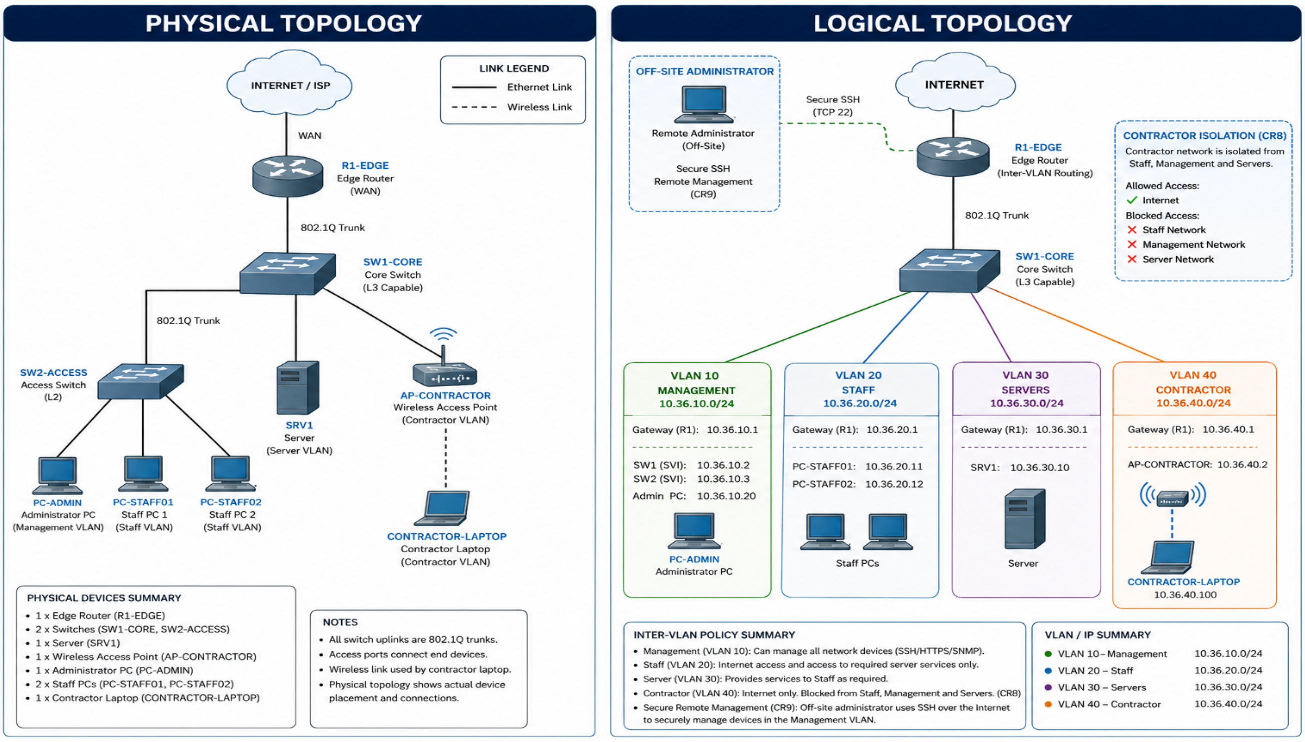

Figure 1. Physical and Logical Topology.

4. VLAN and Logical Network Design

VLAN	Name	Network	Gateway	Purpose
10	MANAGEMENT	10.36.10.0/24	10.36.10.1	Network administration
20	STAFF	10.36.20.0/24	10.36.20.1	Employee workstations
30	SERVERS	10.36.30.0/24	10.36.30.1	Server resources
40	CONTRACTOR	10.36.40.0/24	10.36.40.1	Cleaning/security contractor wireless

5. IP Addressing Plan

Segment	Network	Mask	Usable range	Broadcast	Gateway
VLAN 10	10.36.10.0/24	255.255.255.0	10.36.10.1–10.36.10.254	10.36.10.255	10.36.10.1
VLAN 20	10.36.20.0/24	255.255.255.0	10.36.20.1–10.36.20.254	10.36.20.255	10.36.20.1
VLAN 30	10.36.30.0/24	255.255.255.0	10.36.30.1–10.36.30.254	10.36.30.255	10.36.30.1
VLAN 40	10.36.40.0/24	255.255.255.0	10.36.40.1–10.36.40.254	10.36.40.255	10.36.40.1

6. Security, Contractor Isolation and CR9

• Contractor VLAN 40 is intended to be isolated from Staff VLAN 20, Management VLAN 10, and Server VLAN 30, while allowing the limited external access required by the use case.

• CR9 is addressed through a dedicated management VLAN and secure SSH remote management for the authorised off-site administrator.

• Inter-VLAN access-control rules will be implemented and tested during the implementation stage.

7. Fault-Isolation Challenge

The assigned technical challenge is Network Troubleshooting — Fault Isolation. A controlled configuration fault will be introduced after the network is working. A proposed candidate is an incorrect VLAN assignment on a staff access port. The investigation will use ping, show ip interface brief, show vlan brief, show interfaces trunk, and show running-config, followed by correction and verification.

8. GitHub Repository Plan

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CMPG325-2026-087/
■■■ README.md
■■■ docs/
■■■ diagrams/
■■■ milestone-1/
■■■ packet-tracer/
■■■ configurations/
■■■ testing/
■■■ troubleshooting/
■■■ evidence/
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9. Milestone 1 Checklist

■ Client requirements documented.

■ Physical topology completed.

■ Logical topology completed.

■ IP addressing plan completed.

■ Assigned 10.36.0.0/16 block used.

■ Contractor wireless constraint addressed.

■ CR9 secure remote-management requirement addressed.

■ Assigned fault-isolation challenge documented.

■ Topology diagrams labelled and readable.

■ GitHub repository initialized with meaningful development history.

■ Milestone 1 report included.

■ Design is ready to be implemented in Cisco Packet Tracer after review/approval.

10. Conclusion

The Milestone 1 package documents the client requirements, physical topology, logical topology, IP addressing, security boundaries, CR9, fault-isolation approach, and GitHub structure. It is intended to be the baseline for the subsequent Packet Tracer implementation and testing stages.